SETI – B2 Module – Exam 2021/2022

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TJC (Traffic Jam Chauffeur) system: one of the main components in an automotive driving system

1 Context

We focus on the modeling of a TJC system. The main mission of the TJC is to replace the driver in heavy traffic on the highway: when the speed is less than 70 km/hour, the system automatically follows the vehicle ahead. The TJC manages congestion, assists acceleration and braking, but it also keeps the vehicle in its lane by way of steering interventions. The driver can intervene at any time and override the system. Thus, the main actor of the TJC is the driver; the TJC interacts with several other vehicle systems such as Navigation systems with high precision, Steering and Powertrain systems. Figure 1 and 2 show the actual TJC IHM and dashboard system already embedded by the PSA Group in some of their autonomous under test vehicles.

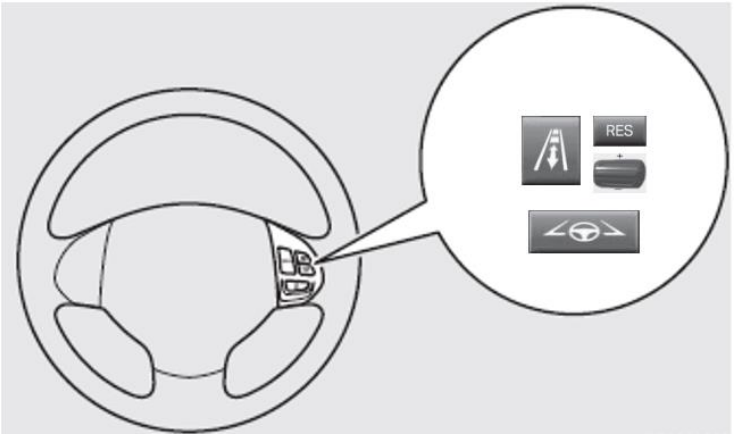

Figure 1: TJC control buttons in the steering wheel

On/Off, activate and deactivate the TJC system

Toggle switch (+/- button): load, change set speed

Restore the last memorized setpoint speed

Adjust the distance between cars

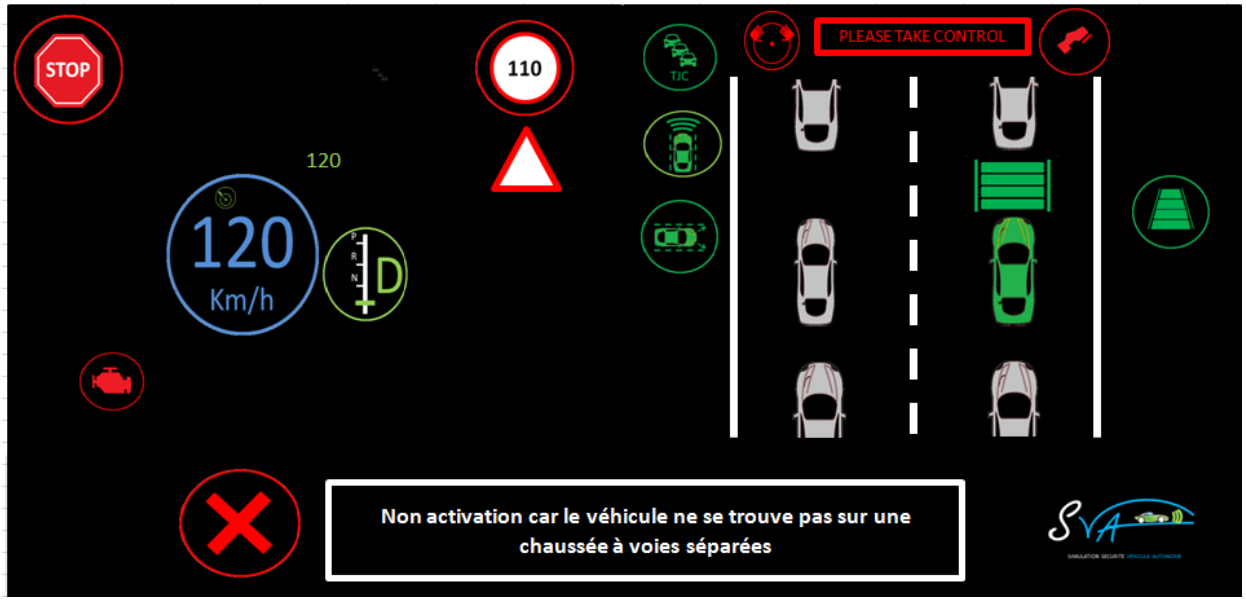

FIGURE 2: DASHBOARD TJC SYSTEM

2 Exercises

2.1 Functions

In a convenient UML Diagram, specify the main System functions as well as its relationships with other components in the autonomous car system or environment (weather, obstacles,...) do not exceed 5 external actors (try to keep the most important actors) to avoid cluttering the diagram

2.2 Composition

In a convenient UML Diagram, specify the composition of the TJC component. You have to follow the composition of an autonomous driving system (perception (Sensors), decision and control (Controller), action (Actuators)). Hence, the TJC component has three sub components: Sensors, Controller and Actuators, which in turn have specific sub-components (e.g. a brake pedal).

The TJC has an activation button (Key ON/OFF) and is connected with the brake and acceleration pedal. It also interacts with the steering wheel and gets perception information. It can act on the engine torque and the steering.

2.3 Behavior

In a convenient UML Diagram, specify the behavior of the TJC component according to its six states outlined below. The Active state is composed of three sub states: Functional, TakeoverRequest and SafetyStop. The TJC moves from the “Unavailable” state to the “StandBy” state only if all activation conditions are available.

We have defined six different operating states for this system based on listed scenarios.

Off

The vehicle is turned off, no TJC functions are available.

Unavailable

The vehicle is turned on, but TJC system is unavailable (disabled) since the conditions for using the system are not met. For example, at start-up, partial wake-up of the computer, etc.:

a) The driver puts the autonomous vehicle into service, the TJC function is not activated and can be activated under certain conditions.
b) The driver requests the activation of the TJC function when the conditions for its use are not met. The TJC function is not activated and the driver is informed of this explicitly (the driver must understand why the system refuses its activation).
c) The TJC function indicates at any time that it is unavailable if the conditions of use are not met.

Standby

The TJC system is ready to be activated by the driver if all conditions of use [cf. R1] are met.

a) The TJC function indicates at any time that it can be activated if the conditions for its use are met.
b) The driver requests the activation of the TJC function when the conditions for its use are met. The TJC function is activated and the driver is informed explicitly.

Active / Functional

When the TJC function is activated by the driver, it keeps the vehicle in its lane and ensures a safe distance from the vehicle in front while guaranteeing passenger comfort.

a) The TJC limits the speed to the allowed maximum speed (from GPS data)
b) If the driver presses the brake pedal or acts on the steering wheel, the TJC function enters the Standby state.
c) If the driver presses the accelerator pedal and the torque demand is greater than the torque setpoint supplied by the TJC, the TJC enters the Standby state.
d) The TJC requires a take-over from the driver, if its operating conditions are no longer met (example when the driver leaves the road with separate lanes or the followed vehicle accelerates to a speed > Vmax TJC).

Active / TakeoverRequired

The TJC function must be able to bring the vehicle to a stop while guaranteeing its safety. The TJC emits a takeover request (e.g. acoustic signal).

a) If the driver acknowledges the request within a given timeout (e.g. 5 seconds), the TJC gets deactivated
b) If the driver does not acknowledge the request within a given time (e.g. he is sleeping or inattentive), the TJC enters a SafetyStop state

Active / SafetyStop

The TJC stops the vehicle in the lane with a deceleration of 2m/s2 (simplified, real systems might have an obstacle detection and adapt braking intensity to avoid a crash) and alerts.

2.4 Domains specific modeling, profile

A simplified automotive profile has four stereotypes. Model the profile (with suitable OCL constraints) and apply the profile to your system model.

a) «ActiveSensor»: an active sensor is represented by an active class having a classifier behavior.
b) «PassiveSensor»: a passive sensor is represented by a normal class. It must contain at least one port with a provided interface
c) «Controller»: a controller is represented by an active class with a classifier behavior. It must have a classifier behavior and attributes indicating the CPU frequency (as an integer with MHz as unit,

alternatively examine MARTE´s value specification language) and the provided memory size (integer in MB or MARTE VSL).

d) «Actuator»: an actuator is represented by a class having at least one port with a provided interface

2.5 Activities

Use a suitable diagram to model the behavior of the speed regulation part. Use the simplified algorithm outlined below.

Obtain the distance to the followed vehicle. If the distance is bigger than an optimal (set) distance, increase the velocity setpoint until the distance is close to it (accelerate). The acceleration is proportional to the difference ($dist_{current}$ - $dist_{set}$), but below a maximum acceleration. If the distance is smaller, reduce it (brake). The brake power is also relative to the difference from the optimal distance ($dist_{set}$ - $dist_{current}$), but increases to a maximum value, if the current distance is smaller than a safe/minimum distance in order to avoid a potential accident.